

Two approaches to identifying fossil pollen

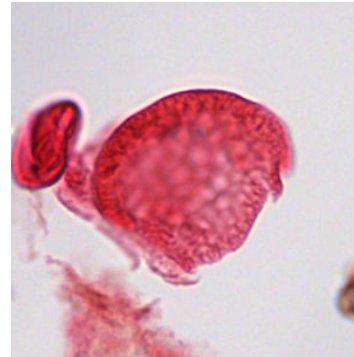
RF-DETR method first • classifier backbones second

01 RF-DETR

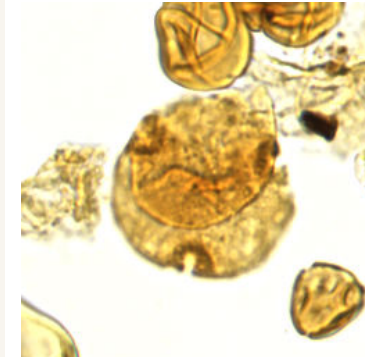
Find specimens and identify categories.

02 Classifier backbones

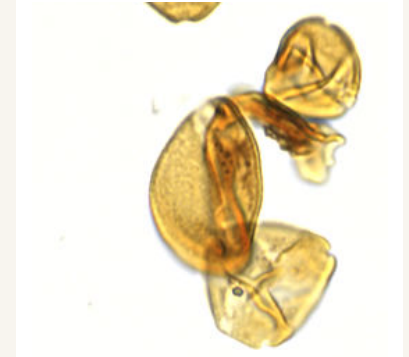
Identify categories from specimen crops.



Bisaccate



Bombacacidites sp.



Arecipites sp.

Classification dataset: 2,283 specimens • 28 broad categories

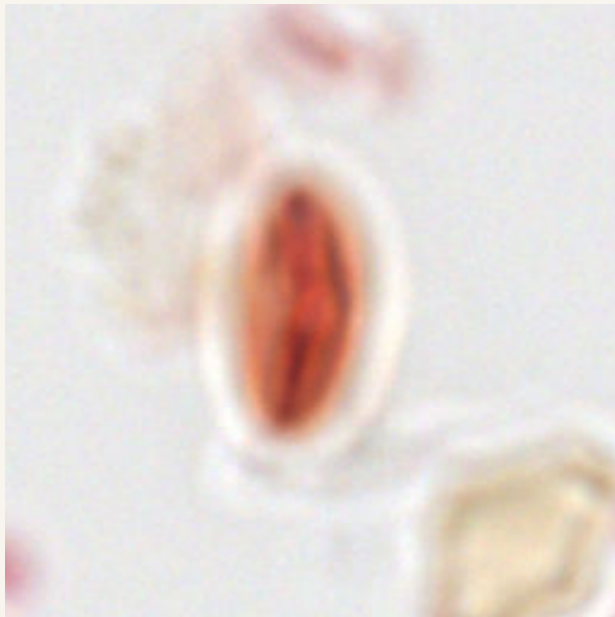
RF-DETR: detection and category identification

RF-DETR team update

Placeholder • content to be added

A clearer view helps the model identify pollen

Original focal plane



Selected sharper plane



Same specimen and crop • expert label: Quercus sp.

Unseen-slide accuracy

32% → 38%

Earlier 32-category trial.
Same Swin settings;
selected plane changed.

Two questions, two ways to check performance

Unseen slides

Can it work on a different slide?

LEARN FROM

Slide A

Slide B

CHECK ON

Slide C

Slide D

Shared slides

Can it identify more grains on familiar slides?

LEARN FROM

A · grain 1

B · grain 1

CHECK ON

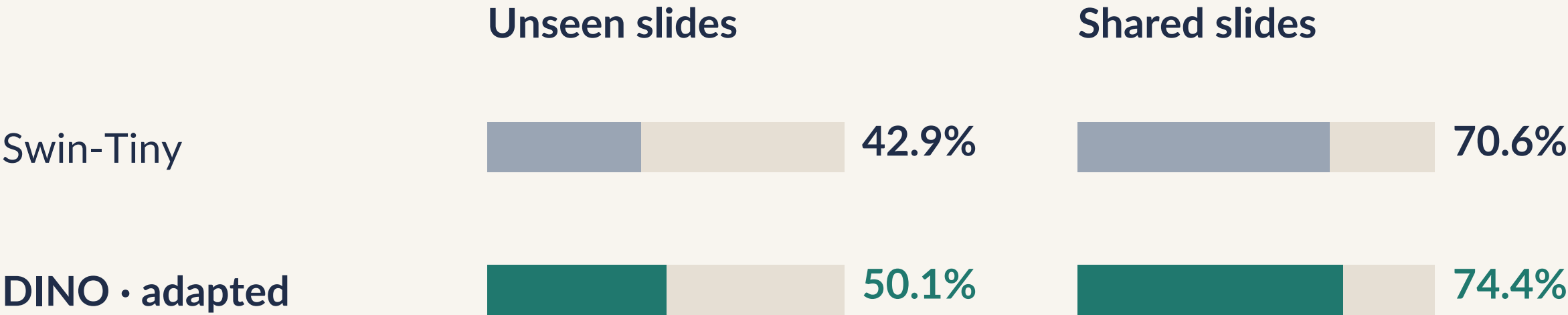
A · grain 2

B · grain 2

Different specimens in every set. Final test slides remain reserved.

DINO leads the broad-category comparison

28 categories · validation accuracy on prepared specimen crops



Grouping makes the task easier; retraining added little DINO accuracy.

Earlier DINO predictions, regrouped: 50.1% unseen / 74.1% shared.

Single-run validation comparisons; final test untouched. These are crop-classification results.

Three categories still need closer review

Adapted DINO · correctly identified / validation examples

CATEGORY	Shared slides	Unseen slides
Bombacacidites sp.	2/7 (29%)	1/17 (6%)
Polyatriopollenites type	2/7 (29%)	1/9 (11%)
Psilatricolpites sp.	3/8 (38%)	3/18 (17%)

Fewer examples may contribute; the main cause is not yet established.

Next: inspect the mistakes, image clarity, and confused categories.

Small validation samples make these percentages uncertain. Momipites is no longer a main weakness.